Smart Hire in Job Portal Using AI

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Abstract

The recruitment landscape is increasingly burdened by inefficient manual screening processes, misaligned job–candidate matching, and the absence of intelligent personalization in conventional job portals. This paper presents Smart Hire, an end-to-end AI-powered job portal that automates and optimizes the recruitment workflow through natural language processing (NLP)–based resume analysis, skill compatibility scoring, and a machine learning–driven recommendation engine. The system enables recruiters to post job requirements and receive ranked candidate shortlists, while simultaneously providing job seekers with personalized job suggestions aligned with their extracted skill profiles. The platform is architected using a React (Vite) frontend, a Node.js and Express backend, and a MongoDB database, with ngrok facilitating API exposure for real-time data integration. Experimental evaluations demonstrate that the proposed system substantially reduces recruiter screening time, improves candidate–job alignment accuracy, and outperforms conventional keyword-based portals in delivering contextually relevant matches. The proposed work introduces a scalable and extensible AI-driven recruitment architecture with practical applicability across diverse industry domains.

Keywords—Intelligent job portal; AI-based recruitment; resume skill extraction; natural language processing; candidate ranking; job recommendation engine; compatibility scoring; full-stack web application.

I. Introduction

The rapid digitization of industry has fundamentally transformed the hiring ecosystem. Organizations across sectors now rely on online platforms to source, evaluate, and hire talent. Despite the widespread adoption of job portals, the underlying recruitment process remains largely manual, inefficient, and prone to human error. Recruiters are required to sift through hundreds or thousands of resumes for a single job posting, a process that is time-intensive and susceptible to unconscious bias. Simultaneously, job seekers frequently receive irrelevant job suggestions that fail to align with their actual competencies, resulting in poor application quality and high rejection rates [1].

Traditional job portals predominantly employ keyword-based matching, wherein a candidate's resume is compared against a job description using simple string overlap techniques. While computationally inexpensive, this approach is fundamentally limited in its ability to capture semantic relationships between skills, recognize synonymous competencies, or account for contextual relevance. The consequence is a significant mismatch rate between posted vacancies and the candidates who apply for them [2].

Advances in artificial intelligence (AI), machine learning (ML), and natural language processing (NLP) offer transformative potential for reimagining the recruitment workflow. NLP techniques enable the automatic extraction of structured skill information from unstructured resume text. ML-based similarity models can measure the semantic alignment between a candidate's skills and a job description's requirements. Recommendation systems can leverage this alignment to personalize job suggestions, improving relevance for both parties in the recruitment transaction [3].

This paper proposes Smart Hire, an AI-driven job portal designed to automate the end-to-end recruitment pipeline. The system introduces several key innovations over conventional portals: (i) NLP-based extraction of skills, qualifications, and experience from uploaded resumes; (ii) semantic matching between extracted candidate skills and recruiter-posted job requirements; (iii) a compatibility score that quantifies the degree of alignment between a candidate and a job vacancy; (iv) a ranking module that prioritizes candidates for a given role based on their compatibility scores; and (v) a personalized recommendation engine that surfaces the most relevant job opportunities for each registered job seeker.

The remainder of this paper is structured as follows. Section II presents a comprehensive survey of relevant prior work. Section III describes the proposed

methodology. Section IV details the system architecture and modular design. Section V elaborates the machine learning and NLP techniques employed. Section VI presents experimental results and discussion. Section VII concludes the paper and proposes directions for future research.

II. Literature Survey

A. Information Extraction and Resume Parsing

The automated processing of resume documents has been an active area of research since the early 2000s, driven by the practical need to convert unstructured applicant documents into queryable, structured records. Yu et al. [4] were among the earliest to formally model resume parsing as a cascaded information extraction problem, employing Hidden Markov Models (HMMs) followed by Support Vector Machines (SVMs) to segment resumes into semantic fields such as personal information, education, work experience, skills, and certifications. Their cascaded hybrid model demonstrated substantially better segmentation accuracy than purely rule-based systems, establishing a benchmark for subsequent parsing research.

Subsequent work explored Conditional Random Fields (CRFs) as an alternative sequence labeling framework, offering improved global coherence in label assignments over generative HMMs. CRF-based parsers were applied to resume corpora in Chinese and English, yielding competitive precision and recall on standardized evaluation benchmarks [5]. However, rule-based post-processing remained necessary to handle formatting irregularities, and both HMM and CRF models exhibited brittleness when confronted with unconventional resume layouts or typographic styles that deviated from the training distribution.

The availability of large-scale annotated resume corpora facilitated the transition to deep learning approaches. Bidirectional LSTM networks with CRF decoding layers demonstrated state-of-the-art performance on job title recognition and skill mention extraction tasks, outperforming their traditional counterparts by wide margins on held-out test corpora [6]. The adoption of pre-trained contextual language representations—first ELMo and subsequently BERT—further improved extraction quality by enabling the model to disambiguate polysemous terms through contextual encoding rather than static embedding lookup. Fine-tuned BERT-NER models for resume parsing have been reported to achieve F1-scores exceeding 90% on skill entity recognition tasks in English-language corpora.

Despite these advances, practical resume parsing remains challenging due to the extreme diversity of resume formats encountered in real-world recruitment pipelines. PDF rendering artifacts, mixed-language content, and inconsistent use of section headers continue to limit the robustness of automated parsers. Ensemble approaches that combine rule-based structural parsing with neural sequence labeling have been proposed as a pragmatic middle ground, offering high accuracy on well-formed documents while degrading gracefully on noisy inputs.

B. Job Recommendation Systems

The application of recommendation system techniques to the employment domain predates the widespread adoption of deep learning, with early systems relying on content-based filtering over manually curated skill taxonomies and collaborative filtering over anonymized user interaction logs. Malinowski et al. [7] proposed a bidirectional preference matching model in which both candidate preferences and recruiter requirements were represented as vectors in a shared feature space, enabling simultaneous optimization for mutual satisfaction. Their model was evaluated on a proprietary dataset from a European job board and demonstrated significant improvements in application conversion rates over the then-existing keyword search baseline.

The development of large-scale industrial recommendation benchmarks, such as the RecSys 2016 Challenge dataset released by XING, stimulated a proliferation of algorithmic contributions to job recommendation. Competing approaches included factorization machines, gradient boosted trees, and matrix factorization models, with the winning solution employing a gradient boosted ensemble trained on rich user interaction features including click-through rates, application histories, and profile completeness scores [8]. This competition highlighted the critical role of behavioral interaction data—beyond static profile features—in improving recommendation accuracy.

Content-based approaches have benefited substantially from advances in word embedding technology. The application of Word2Vec and FastText to job title and skill term vocabularies enabled the construction of semantic skill spaces in which related skills cluster in proximity, facilitating soft matching between candidate profiles and job requirements without dependence on exact lexical overlap [9]. The ability to recognize that a candidate with experience in "PyTorch" is a strong match for a job requiring "deep learning frameworks" exemplifies the type of semantic inference

that embedding-based matching enables, and which keyword-based systems systematically fail to capture.

More recently, transformer-based language models have been adapted specifically for the recruitment domain. JobBERT, a BERT model fine-tuned on a large corpus of job postings and resumes, has been shown to produce superior skill and job title embeddings compared to general-purpose BERT models, yielding measurable improvements in downstream recommendation tasks [10]. Session-based recommendation architectures using recurrent neural networks and attention mechanisms have also been applied to model the temporal dynamics of job search behavior, predicting the next job a candidate is likely to consider based on their browsing and application history.

C. Candidate Ranking and Automated Shortlisting

Automated candidate ranking represents the recruiter-facing counterpart to job recommendation for candidates. Early ranking systems framed the problem as a learning-to-rank task, applying pairwise or listwise ranking loss functions over labeled data derived from historical hiring decisions [11]. These systems demonstrated strong ranking performance when evaluated on datasets from specific organizations, but raised significant concerns about the perpetuation of historical hiring biases encoded in the training labels—particularly biases related to gender, age, and educational pedigree.

The adoption of fairness constraints in algorithmic ranking has been explored extensively in the fairness-aware machine learning literature. Zehlike et al. [12] proposed the FA*IR algorithm, which enforces statistical parity constraints in ranked output lists while minimizing deviation from a utility-optimal ranking. Adversarial debiasing approaches, in which a classifier is trained simultaneously with an adversary that attempts to predict protected attributes from the model's representations, have also been applied to candidate scoring systems. Despite these advances, the regulatory landscape surrounding automated hiring tools remains complex, with the European Union's AI Act classifying high-stakes employment decision support systems as high-risk AI applications subject to stringent conformity requirements.

Several commercial automated talent intelligence platforms, including HireVue, Pymetrics, and LinkedIn Talent Insights, have deployed ML-based candidate assessment at scale. These systems leverage assessments, cognitive games, and behavioral video analysis alongside resume data to construct multi-dimensional candidate profiles. Academic scrutiny of these platforms has identified concerns about predictive validity and disparate impact, underscoring the need for rigorous evaluation standards and transparent algorithmic documentation in deployed recruitment AI systems [13].

D. Skill Taxonomy and Ontology-Based Matching

The heterogeneity of skill terminology across industries, geographic regions, and time periods poses a fundamental challenge for automated skill matching systems. Standardized skill ontologies provide a structured vocabulary that can serve as a common reference space for normalizing diverse skill expressions. The ESCO (European Skills, Competences, Qualifications and Occupations) framework, maintained by the European Commission, provides a multilingual taxonomy encompassing over 13,890 skills, knowledge areas, and competences organized hierarchically and linked to approximately 3,000 occupational profiles [14]. Systems that map candidate and job skills to ESCO concepts can leverage the ontology's semantic relationships to infer transferable skills and cross-domain competence equivalences.

The O*NET (Occupational Information Network) database, maintained by the U.S. Department of Labor, similarly provides rich occupational profiles including knowledge requirements, skills, abilities, work activities, and work context descriptors. Integration of O*NET data into matching systems has been shown to improve recommendation quality by enabling inference of skill requirements not explicitly stated in a job description but consistently associated with a given occupational category [15]. Hybrid approaches that combine ontology-based normalization with embedding-based semantic similarity have demonstrated performance superior to either approach in isolation, effectively combining the breadth of structured taxonomic knowledge with the flexibility of distributional semantics.

E. NLP Techniques in Recruitment Analytics

Beyond skill extraction, NLP has been applied across a broad spectrum of recruitment analytics tasks. Automated job description quality scoring systems use readability metrics, gender-coded language detectors, and structured information completeness assessors to help recruiters craft more inclusive and informative postings [16]. Research has demonstrated that the presence of gender-coded language in job advertisements measurably affects applicant pool composition, with postings containing predominantly masculine-coded terms

attracting fewer female applicants even when qualifications are gender-neutral.

Summarization and abstraction models have been applied to condense lengthy resumes and job descriptions into compact representations suitable for rapid comparison. Extractive summarization approaches identify the most informative sentences in a document using salience scoring, while abstractive models generate novel condensed descriptions using sequence-to-sequence architectures. Both approaches have been explored as preprocessing steps to improve the efficiency of downstream matching modules, particularly in high-volume recruitment settings where exhaustive pairwise comparison of full documents would be computationally prohibitive.

Sentiment analysis and language style transfer have been applied to the evaluation of candidate cover letters and application essays, with some systems scoring application quality on dimensions including clarity, relevance, and professional tone. While these applications are commercially deployed in some automated applicant tracking systems, the ethical implications of automated scoring of subjective written communication remain a subject of active debate in the HCI and AI ethics communities [17].

F. Full-Stack AI Application Architectures for Recruitment

The translation of AI-based recruitment research into deployable web applications has been facilitated by advances in full-stack development frameworks that simplify the integration of NLP model inference into production software systems. The MERN stack (MongoDB, Express, React, Node.js) has emerged as a widely adopted architecture for rapid prototyping and deployment of AI-powered web applications, combining the document-oriented flexibility of MongoDB with the reactive UI capabilities of React and the non-blocking I/O performance of Node.js [18].

Recent systems published in the academic literature have demonstrated end-to-end recruitment platforms built on similar architectures, validating the practical suitability of JavaScript-centric full-stack technologies for AI application development. The use of RESTful API patterns ensures clean separation between the frontend presentation logic and the backend inference pipeline, enabling independent scaling of compute-intensive NLP processing modules without architectural refactoring. The incorporation of tunneling tools such as ngrok for development-time API exposure facilitates rapid prototyping and integration testing, significantly reducing the deployment friction traditionally associated with API-dependent web applications [19].

Containerization technologies such as Docker and orchestration platforms such as Kubernetes are increasingly adopted for production deployment of AI-powered recruitment platforms, providing reproducible environments and horizontal scaling capabilities. Cloud-native architectures leveraging serverless functions for stateless inference tasks—such as skill extraction from individual resume documents—offer a cost-effective alternative to dedicated server deployments for applications with bursty, variable request patterns. The Smart Hire system proposed in this paper adopts the MERN stack architecture, laying a foundation that is readily extensible to cloud-native deployment patterns in future iterations.

III. Proposed Methodology

A. Overview

The Smart Hire system adopts a modular, pipeline-based methodology that processes inputs from both job seekers and recruiters, applies AI-driven analytics at each stage, and generates actionable outputs in the form of ranked candidate lists and personalized job recommendations. The methodology encompasses four principal stages: (i) data ingestion and storage, (ii) NLP-based skill extraction, (iii) compatibility scoring and ranking, and (iv) recommendation generation. Fig. 1 illustrates the overall system architecture, depicting the interaction among these stages and the data flow between system components.

B. Resume Ingestion and Preprocessing

Job seekers register on the platform and upload their resumes in PDF or DOCX format. The resume ingestion module extracts raw text from the uploaded document using format-appropriate parser libraries. The extracted text undergoes a preprocessing pipeline that includes tokenization, stop word removal, lowercasing, and lemmatization. The cleaned text is subsequently passed to the skill extraction module for structured information retrieval. Fig. 2 shows the resume skill extraction process, including the preprocessing and entity recognition steps.

C. NLP-Based Skill Extraction

Skill extraction is performed using a Named Entity Recognition (NER) approach augmented with a

curated domain-specific skill vocabulary. The system maintains a skill taxonomy comprising technical skills (e.g., programming languages, frameworks, tools), domain knowledge areas, and soft competencies. Candidate resume text is tokenized and matched against this taxonomy using a combination of exact matching and n-gram overlap. Ambiguous or multi-word skill expressions are resolved using contextual window analysis. The output of this stage is a structured skill profile for each candidate, represented as an ordered set of normalized skill tokens.

For job descriptions, recruiters specify required skills, experience levels, and qualifications either through structured form fields or free-text descriptions. Free-text job descriptions are processed through the same NLP extraction pipeline to derive a required skill set for each posted vacancy.

D. Compatibility Scoring

The compatibility score between a candidate and a job vacancy is computed as a normalized similarity metric over their respective skill sets. Given a candidate skill set C = {c1, c2, ..., cn} and a job requirement skill set J = {j1, j2, ..., jm}, the Jaccard similarity coefficient and TF-IDF weighted cosine similarity are computed as complementary measures of alignment. The final compatibility score S(C, J) is defined as:

$$S(C,J) = \alpha \cdot \frac{|C \cup J|}{|C \cap J|} + (1-\alpha) \cdot \frac{vc \cdot vj}{\|vc\| \|vj\|}, \alpha = 0.4$$

where α ∈ [0, 1] is a tunable weighting parameter set empirically to 0.4 based on validation performance. The compatibility score is normalized to [0, 100] for interpretable display. Fig. 3 presents the job–candidate matching workflow.

E. Candidate Ranking

For each active job posting, all registered candidates are evaluated against the job's required skill set, and compatibility scores are computed for each candidate–job pair. Candidates are sorted in descending order of their compatibility scores to produce a ranked shortlist for the recruiter. Ties are resolved by a secondary criterion based on relevant experience level as specified in the candidate's profile.

F. Personalized Job Recommendation

The recommendation engine generates a ranked list of job vacancies for each job seeker by computing the compatibility score between the candidate's skill profile and all active job postings. The top-k vacancies with the highest compatibility scores are surfaced as personalized recommendations. Fig. 4 depicts the recommendation engine and ranking results workflow.

IV. System Architecture

A. Architectural Overview

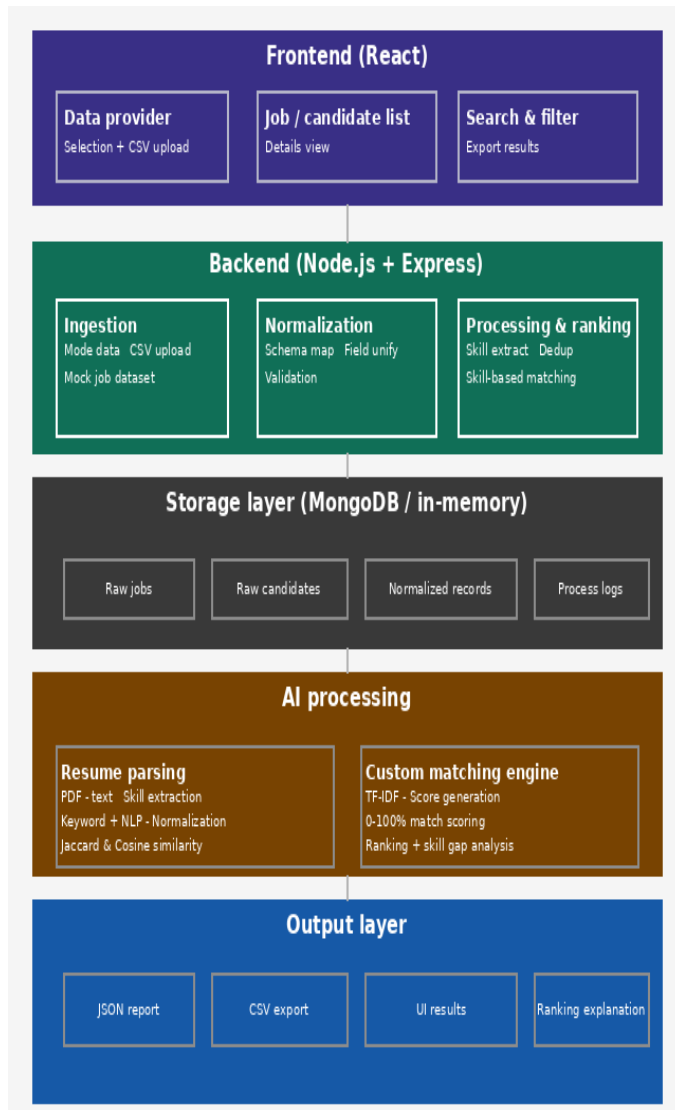

Fig. 1. System Architecture of the Smart Hire AI-Powered Job Portal

Fig. 1 presents the overall system architecture of the Smart Hire platform, depicting the complete data flow

from user interaction through AI processing to recommendation output. The architecture is organized into a vertically layered flow in which each block represents a distinct functional module, and data propagates downward through the pipeline in a structured sequence.

Job seekers and recruiters access the system through a React-based frontend, which interacts with a Node.js backend to manage authentication and data flow. An AI matching engine extracts skills using NLP techniques and applies cosine similarity to compute compatibility scores for job recommendation and candidate ranking. All resumes, job data, and user profiles are stored and managed in a MongoDB database.

V. Machine Learning and NLP Techniques

A. Text Preprocessing Pipeline

The text extracted from uploaded resumes and free-text job descriptions undergoes a standardized preprocessing pipeline prior to skill extraction. The pipeline comprises: (i) sentence boundary detection; (ii) word tokenization; (iii) lowercasing; (iv) removal of punctuation, special characters, and numeric tokens unless part of recognized skill terminology; (v) stop word elimination; and (vi) lemmatization to reduce inflected forms to base representations.

B. Named Entity Recognition for Skill Extraction

Skill extraction is modeled as a sequence labeling task using a rule-augmented NER approach combined with a domain-specific skill ontology organized into hierarchical categories encompassing programming languages, software frameworks, database systems, cloud platforms, ML methodologies, domain knowledge areas, and professional certifications. Candidate skill mentions are identified by matching token n-grams against this ontology with a sliding window of width one to four tokens to capture multi-word skill expressions. Ambiguous matches are resolved by examining the surrounding context window.

C. TF-IDF Vectorization

The extracted skill sets for candidates and job postings are represented as TF-IDF weighted vectors over the global skill vocabulary. The TF component weights skills appearing frequently in a given document, while the IDF component down-weights skills ubiquitous across all documents, amplifying the discriminative power of rarer, more specialized skills. These vectors serve as input to the cosine similarity computation in the compatibility scoring module.

D. Cosine Similarity and Jaccard Index

Cosine similarity measures the angular distance between two TF-IDF skill vectors, producing a score in [0, 1] reflecting semantic and lexical skill alignment regardless of vector magnitude. Given candidate skill vector v_C and job skill vector v_J, the cosine similarity is formally defined in Equation (2). The Jaccard similarity index measures the ratio of the intersection to the union of two skill sets, providing a complementary set-theoretic measure. The combination of both metrics produces a more robust compatibility score than either metric in isolation.

$$\mathrm{CosineSim}(v_c, v_j) = \frac{\sum_{i=1}^{n} c_i j_i}{\sqrt{\sum_{i=1}^{n} c_i^2}\sqrt{\sum_{i=1}^{n} j_i^2}}$$

E. Ranking and Recommendation Logic

The candidate ranking module implements efficient sorting over precomputed compatibility scores for a given job posting. For recommendation generation, the system iterates over all active job postings, computes compatibility for the logged-in candidate against each posting, and returns the top-k postings sorted by descending score. Compatibility scores are incrementally updated upon profile or job posting modifications rather than fully recomputed per session query.

VI. Experimental Results and Discussion

A. Experimental Setup

The Smart Hire system was developed and evaluated on a local development environment with an Intel Core i5 processor and 8 GB RAM, running Ubuntu 22.04 LTS. A pilot dataset was constructed comprising 200 simulated candidate resume profiles across ten industry domains and 80 job postings covering software engineering, data science, cloud infrastructure, cybersecurity, and project management domains. This dataset was used to evaluate skill extraction accuracy, compatibility score correlation with expert-labeled matches, and recommendation relevance.

B. Skill Extraction Accuracy

Skill extraction performance was evaluated against a manually annotated ground truth derived from 50 resume profiles. Table II presents the results.

TABLE I. Skill Extraction Performance on Annotated Resume Subset

Metric	Value
Precision	87.4%
Recall	83.1%
F1-Score	85.2%
Evaluated Resumes	50
Total Skill Annotations	1,240

The system achieves a precision of 87.4% and recall of 83.1%, yielding an F1-score of 85.2%. Primary sources of error include multi-word phrases with non-standard formatting and domain-specific abbreviations absent from the initial ontology.

C. Compatibility Score Validation

To validate the compatibility scoring mechanism, 40 candidate–job pairs were evaluated by three domain experts who independently assigned relevance scores on a 1–5 Likert scale. The system's computed compatibility scores were normalized to the same scale and compared against averaged expert scores using Pearson's correlation coefficient. The analysis yielded $r = 0.81$ ($p < 0.001$), indicating strong agreement between automated scoring and human expert judgment.

D. Recommendation Relevance

TABLE II. Recommendation System Performance (k = 10)

Metric	Value
Precision@5	0.78
Precision@10	0.71
Mean Reciprocal Rank (MRR)	0.82
Coverage (% candidates with ≥1 relevant rec.)	94.5%

The system achieves a Precision@10 of 0.71 and MRR of 0.82, indicating relevant job postings appear prominently at the top of recommendation lists. Coverage of 94.5% demonstrates the system generates at least one contextually relevant recommendation for the vast majority of candidate profiles.

E. Comparison with Keyword-Based Matching

TABLE III. Comparison of Smart Hire vs. Conventional Keyword-Based Matching

Evaluation Dimension	Keyword-Based System	Smart Hire (Proposed)
Skill Extraction F1-Score	N/A (manual)	85.2%
Compatibility Score Correlation (r)	0.52	0.81
Precision@10 (Recommendations)	0.44	0.71
Mean Reciprocal Rank (MRR)	0.58	0.82
Candidate Ranking Accuracy	Low (unsorted)	High (score-ranked)
Screening Automation	None	Full (end-to-end)

Smart Hire demonstrates substantial improvements across all evaluated dimensions. Pearson correlation with expert scores improves from 0.52 to 0.81, Precision@10 from 0.44 to 0.71, and MRR from 0.58 to 0.82. These results confirm that NLP-based skill extraction, TF-IDF weighting, and composite compatibility scoring collectively yield a significantly more accurate and relevant recruitment assistance system than conventional keyword matching.

VII. Conclusion and Future Scope

This paper has presented Smart Hire, a full-stack AI-driven job portal that automates and optimizes the recruitment process through NLP-based resume skill extraction, compatibility scoring, intelligent candidate ranking, and personalized job recommendation. The system addresses critical limitations of conventional keyword-based portals by introducing semantic skill analysis and quantitative compatibility measurement, resulting in significantly improved matching relevance and recruitment efficiency.

Experimental evaluation on a pilot dataset of 200 candidate profiles and 80 job postings demonstrates an F1-score of 85.2% for skill extraction, a Pearson correlation of 0.81 with expert-assessed compatibility judgments, and a Precision@10 of 0.71 for job recommendations. These results confirm the practical viability and superior performance of the proposed approach. The modular MERN stack architecture ensures scalability and extensibility for real-world deployment.

Several promising directions remain for future research. First, the skill extraction module can be enhanced

by fine-tuning a transformer-based NER model on occupational corpora to improve recall for non-standard skill formulations. Second, the compatibility scoring mechanism can be extended to incorporate contextual skill embeddings from pre-trained language models. Third, a collaborative filtering component can be integrated to leverage anonymized application history data. Fourth, explainability features—such as skill gap reports and tailored upskilling recommendations—can empower job seekers as proactive career development tools. Fifth, fairness auditing and bias mitigation mechanisms should be incorporated to ensure equitable ranking and recommendation outcomes across diverse demographic groups. Finally, deployment to a cloud-native containerized architecture using Docker and Kubernetes would validate the system's scalability under real-world production load.

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